

① Processos Estocásticos (Dia 28/09/2026)

~~Marting~~ Comentário: Aula passada: Tempo de parada. Demonstração $X_n = X_{\wedge n}$.

Convergência de Martingais

Teo: $(M_n)_{n \in \mathbb{N}}$ martingal, $\mathbb{E}(M_n^2) \leq C \quad \forall n \in \mathbb{N}$

$\Rightarrow \exists M_\infty$ t.q. ~~$\mathbb{E}(M_\infty^2) \leq C$~~ $\mathbb{E}(M_\infty^2) \leq C$ e $M_n \rightarrow M_\infty$ q.c e em L^2 .

prova: Fixe $m \in \mathbb{N}$.

- $X_k := (M_{k+m} - M_m)^2$

- $(X_k)_{k \in \mathbb{N}}$ é submartingal

- Desigualdade de Doob $\Rightarrow \mathbb{P}\left(\max_{0 \leq k \leq n} X_k \geq \lambda\right) \leq \frac{\mathbb{E}(X_n)}{\lambda}$

- Seja $d_k = M_k - M_{k-1}$

- $\mathbb{E}(X_n) = \mathbb{E}((M_{n+m} - M_m)^2) = \mathbb{E}\left(\left(\sum_{k=m+1}^{n+m} d_k\right)^2\right)$

- $\mathbb{E}(d_k d_j) = \mathbb{E}(\mathbb{E}(d_k d_j | \mathcal{F}_k)) = \mathbb{E}(d_k \underbrace{\mathbb{E}(d_j | \mathcal{F}_k)}_{=0}) = 0$

$$\begin{aligned} \Rightarrow \mathbb{E}(X_n) &= \mathbb{E}\left(\left(\sum_{k=m+1}^{n+m} d_k\right)^2\right) = \sum_{k=0}^n \mathbb{E}(d_k^2) \\ &\leq \sum_{k=0}^{\infty} \mathbb{E}(d_k^2) \quad (*) \end{aligned}$$

$$(2) \cdot M_{-1} = 0$$

$$\Rightarrow C \geq \mathbb{E}(M_n^2) = \sum_{k=0}^n \mathbb{E}(d_k^2) \Rightarrow \sum_{k=0}^n \mathbb{E}(d_k^2) \leq C \quad \forall n \in \mathbb{N}.$$

$$\Rightarrow \sum_{k=0}^{\infty} \mathbb{E}(d_k^2) \leq C.$$

$$\Rightarrow \mathbb{P}\left(\sup_{k \geq 0} |M_{k+m} - M_m| > \varepsilon\right)$$

$$= \mathbb{P}\left(\bigcup_{n \geq 0} \left\{ \max_{0 \leq k \leq n} |M_{k+m} - M_m| > \varepsilon \right\}\right)$$

$$= \lim_{n \rightarrow \infty} \mathbb{P}\left(\max_{0 \leq k \leq n} |M_{k+m} - M_m| > \varepsilon\right)$$

$$= \lim_{n \rightarrow \infty} \mathbb{P}\left(\sup_{k \leq n} X_k \geq \varepsilon^2\right) \leq \lim_{n \rightarrow \infty} \frac{\mathbb{E}(X_n)}{\varepsilon^2} \stackrel{(*)}{\leq} \frac{1}{\varepsilon^2} \sum_{k=m+1}^{\infty} \mathbb{E}(d_k^2) \stackrel{(\Delta)}{\leq} \frac{1}{\varepsilon^2} C$$

Markov

• Para cada $i \in \mathbb{N}$, escolha $n_i \in \mathbb{N}$ tal que $\sum_{k=n_i+1}^{\infty} \mathbb{E}(d_k^2) \leq 8^{-i}$ (□)

$$\Rightarrow \mathbb{P}\left(\sup_{k \geq 0} |M_{k+n_i} - M_{n_i}| \geq 2^{-i}\right) \stackrel{\Delta+1}{\leq} 2^{-i}$$

• Seja $A_i = \left\{ \sup_{k \geq 0} |M_{k+n_i} - M_{n_i}| \geq 2^{-i} \right\}$

• $\sum_{i=1}^{\infty} \mathbb{P}(A_i) < \infty$

③ Borel-Cantelli $\Rightarrow \mathbb{P}(A_i \text{ i.o.}) = 0$.

$\Rightarrow$ q.c. $\exists i \in \mathbb{N}$ t.q. $\forall j \geq i, \sup_{K \geq 0} |M_{K+n_j} - M_{n_j}| < 2^{-j}$

$\Rightarrow$ q.c. $\exists i$ t.q. ~~$\forall j \geq i$~~ , $|M_{K+n_j} - M_{n_j}| < 2^{-j} \quad \forall K \geq 0$

$\Rightarrow$ q.c. $(M_n)_{n \in \mathbb{N}}$ é Cauchy

$\Rightarrow \exists M_\infty : M_n \rightarrow M_\infty$ q.c.

$$M_\infty = \sum_{K=0}^{\infty} d_K$$

$$\Rightarrow \mathbb{E}((M_\infty - M_n)^2) = \sum_{K=n+1}^{\infty} \mathbb{E}(d_K^2) \xrightarrow{n \rightarrow \infty} 0 \Rightarrow M_n \rightarrow M_\infty \text{ em } L^2$$

Def : $(M_n)_{n \in \mathbb{N}}$ submartingal • $a < b \in \mathbb{R}$.

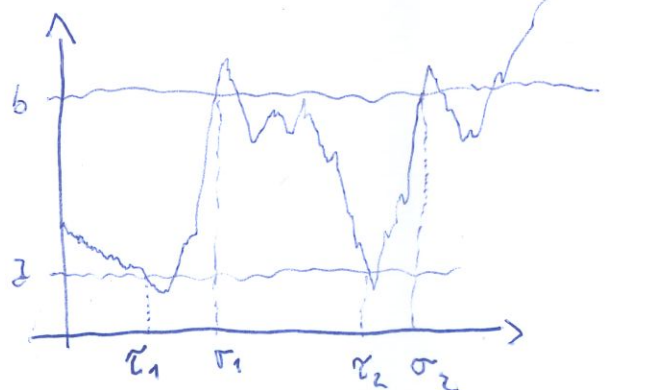
$$\tau_0 = \sigma_0 = 0$$

$$\tau_K = \inf \{ n \geq \sigma_K : M_n \leq a \} \quad \forall K \in \mathbb{N}$$

$$\sigma_K = \inf \{ n \geq \tau_K : M_n \geq b \} \quad \forall K \in \mathbb{N}$$

$$N_n(a, b) = \sum_{K=1}^{\infty} \mathbb{1}_{\{\sigma_K \leq n\}}$$

$N_n(a, b)$ é o "número de up-crossing"



4)

Prop: $(M_n)_{n \in \mathbb{N}}$ submartingal $\Rightarrow \mathbb{E}(N_n(a,b)) \leq \frac{\mathbb{E}((M_n - a)^+)}{b-a}$

onde $(x-a)^+ = \max\{x-a, 0\}$.

prova: $((M_n - a)^+)$ é submartingal (porque $x \mapsto (x-a)^+$ é convexa crescente)

• "Estratégia": $A_{n+1} = \mathbb{1}_{\{\exists k: \tau_k \leq n < \sigma_k\}}$

• $(A_n)_{n \in \mathbb{N}}$ é $\tilde{n}$ -antecipativo e $A_n \in \{0, 1\}$

~~$\Rightarrow ((A \cdot (M-a)^+)_n)_{n \in \mathbb{N}}$ é submartingal~~

$\Rightarrow \mathbb{E}[(A \cdot (M-a)^+)_n] \leq \mathbb{E}((M_n - a)^+)$. (Prop 2.2.11)

• $(b-a)N_n(a,b) \leq (A \cdot (M-a)^+)_n$

$\Rightarrow \mathbb{E}((b-a)N_n(a,b)) \leq \mathbb{E}[(A \cdot (M-a)^+)_n] \leq \mathbb{E}((M_n - a)^+)$

Teo: ~~$(M_n)_{n \in \mathbb{N}}$~~ $(M_n)_{n \in \mathbb{N}}$ martingal, $\mathbb{E}(|M_n|) \leq C \quad \forall n \in \mathbb{N}$.

$\Rightarrow \exists M_\infty$, $\mathbb{E}(|M_\infty|) \leq C$ t.q. $M_n \rightarrow M_\infty$ a.s.

⑤ prova: Sejam $a, b \in \mathbb{Q}$, $a < b$, ~~A_{ab}~~ .

$$A_{ab} = \left\{ \liminf_{n \rightarrow \infty} M_n \leq a < b < \limsup_{n \rightarrow \infty} M_n \right\}$$

• " $(M_n)_{n \in \mathbb{N}}$ converge q.c." $\iff$ " $P(A_{ab}) = 0 \ \forall a < b$ em $\mathbb{Q}$."

• $(N_n(a, b))_{n \in \mathbb{N}}$ é monotônica $\Rightarrow \exists N_\infty(a, b)$ em $\mathbb{N} \cup \{+\infty\}$ t.q.

$$N_n(a, b) \longrightarrow N_\infty(a, b) \text{ q.c.}$$

• $A_{ab} \subseteq \{N_\infty(a, b) = \infty\}$ Quero $P(N_\infty(a, b) = \infty) = 0$

$$\mathbb{E}(N_n(a, b)) \overset{\text{up-crossing prop.}}{\leq} \frac{1}{b-a} \mathbb{E}((M_n - a)^+) \leq \frac{1}{b-a} \mathbb{E}|M_n - a|$$

$$\leq \frac{1}{b-a} (|a| + \mathbb{E}|M_n|) \leq \frac{|a| + C}{b-a}$$

$$\bullet \text{ TCM} \Rightarrow \mathbb{E}(N_\infty(a, b)) = \lim_{n \rightarrow \infty} \mathbb{E}(N_n(a, b)) \leq \frac{|a| + C}{b-a} < \infty$$

$$\Rightarrow P(N_\infty(a, b) = \infty) = 0$$

$$\Rightarrow M_n \longrightarrow M_\infty$$

$$\text{Fatou} \Rightarrow \mathbb{E}|M_\infty| \leq \liminf_{n \rightarrow \infty} \mathbb{E}|M_n| \leq C$$

$$\Rightarrow M_\infty \in L^1$$

~~11~~

⑥ Cor: $(S_n)_{n \in \mathbb{N}}$ passeio aleatório simples simétrico em $\mathbb{Z}$

$$\tau = \inf\{n: S_n = A \text{ ou } S_n = -B\} \rightarrow \mathbb{P}(\tau < \infty) = 1.$$

prova: $M_n = S_n \wedge \tau$ é martingal.

$$\forall \omega \in \{\tau = \infty\}, |M_{n+1}(\omega) - M_n(\omega)| = |S_{n+1}(\omega) - S_n(\omega)| = 1 \quad \forall n.$$

$\Rightarrow \forall \omega \in \{\tau = \infty\}, (M_n)_{n \in \mathbb{N}}$ não converge.

• $(M_n)_{n \in \mathbb{N}}$ é martingal limitado

$$\Rightarrow M_n \rightarrow M_\infty \text{ q.c.}$$

$$\Rightarrow \mathbb{P}(\tau < \infty) = 0.$$

Exemplo: $(X_n)_{n \in \mathbb{N}}$ iid. $\mathbb{P}(X_n = 0) = \mathbb{P}(X_n = 2) = 1/2$.

$$\bullet M_n = \prod_{k=1}^n X_k$$

• $(M_n)_{n \in \mathbb{N}}$ é martingal

$$\bullet \mathbb{E}|M_n| = 1.$$

$$\Rightarrow M_n \rightarrow M_\infty \text{ q.c.}$$

$$\bullet M_\infty = 0$$

Observe que $\mathbb{E}(M_\infty) = 0 \neq 1 = \mathbb{E}(M_n)$

$$\Rightarrow M_n \not\rightarrow M_\infty \text{ em } L^1.$$